Graphing Rational Functions

ANSWER KEY

Reading graphs

1 The graph of $f(x) = \frac{x+2}{x-3}$ is shown. From the graph and the formula, state the asymptotes and intercepts.

Vertical asymptote $x = 3$; horizontal asymptote $y = 1$; intercepts $(-2, 0)$ and $\left(0, -\frac{2}{3}\right)$.

2 The function $g(x) = \frac{1}{x-2} + 3$ is a transformation of $y = \frac{1}{x}$.

a Describe the transformation. Shift right 2 units and up 3 units.

b State the asymptotes of g. Vertical $x = 2$; horizontal $y = 3$.

c State the domain and range. Domain $x \neq 2$; range $y \neq 3$.

Sign analysis

3 Let $f(x) = \frac{x-1}{(x+2)(x-4)}$. Using a sign chart with critical values $x = -2, 1, 4$, determine the intervals on which $f(x) > 0$.

Testing each interval: $(-\infty, -2)$ negative; $(-2, 1)$ positive; $(1, 4)$ negative; $(4, \infty)$ positive. So $f(x) > 0$ on $(-2, 1) \cup (4, \infty)$.

4 Solve the rational inequality $\frac{x+3}{x-2} \leq 0$.

Critical values $x = -3$ (zero) and $x = 2$ (undefined). The quotient is negative between them: $[-3, 2)$ — including -3, excluding 2.

5 Sketch-planning: for $h(x) = \frac{2x}{x^2-1}$, state the asymptotes and the symmetry of the graph.

Vertical asymptotes $x = 1$ and $x = -1$; horizontal asymptote $y = 0$. Since $h(-x) = -h(x)$, the function is odd — symmetric about the origin.

Sketching a complete graph

6 For $f(x) = \frac{x^2-1}{x^2-4}$, find all intercepts and asymptotes, then describe the graph's behaviour near each vertical asymptote.

x-intercepts: $x = \pm 1$. y-intercept: $f(0) = \frac{-1}{-4} = \frac{1}{4}$. Vertical asymptotes $x = \pm 2$; horizontal asymptote $y = 1$ (equal degrees). Near $x = 2$: $f \to +\infty$ as $x \to 2^+$ and $f \to -\infty$ as $x \to 2^-$. Near $x = -2$: by symmetry (the function is even), $f \to -\infty$ as $x \to -2^+$ and $f \to +\infty$ as $x \to -2^-$.

7 Sketch-planning: the function $g(x) = \frac{3}{x-1} - 2$ has a vertical asymptote at $x = 1$ and horizontal asymptote $y = -2$. Find its x- and y-intercepts.

y-intercept: $g(0) = \frac{3}{-1} - 2 = -5$. x-intercept: $0 = \frac{3}{x-1} - 2$ gives $\frac{3}{x-1} = 2$, so $x - 1 = \frac{3}{2}$, $x = \frac{5}{2}$.

More rational inequalities

8 Solve $\frac{(x-2)(x+1)}{x-4} \geq 0$.

Critical values $x = -1, 2, 4$. Sign chart: negative on $(-\infty, -1)$, positive on $(-1, 2)$, negative on $(2, 4)$, positive on $(4, \infty)$. Solution (including zeros, excluding the undefined point): $[-1, 2] \cup (4, \infty)$.

9 Solve $\frac{2}{x+3} > 1$.

Rewrite: $\frac{2}{x+3} - 1 > 0 \Rightarrow \frac{2-(x+3)}{x+3} > 0 \Rightarrow \frac{-x-1}{x+3} > 0 \Rightarrow \frac{x+1}{x+3} < 0$. Critical values $-3, -1$: solution $(-3, -1)$.

Building a rational function from a description

10 Construct a rational function with vertical asymptote $x = 2$, horizontal asymptote $y = 1$, and a hole at $x = -3$.

Start with a base of $\frac{x}{x-2}$ (horizontal asymptote $y = 1$, vertical asymptote $x = 2$), then introduce a common factor $(x+3)$ in both numerator and denominator: $f(x) = \frac{x(x+3)}{(x-2)(x+3)}$.

Graphing with a slant asymptote

11 For $f(x) = \frac{x^2-4}{x-1}$, find the slant asymptote and the vertical asymptote, and use them together with intercepts to describe the overall shape of the graph.

Dividing: $x^2 - 4 = (x-1)(x+1) - 3$, so $f(x) = x + 1 - \frac{3}{x-1}$. Slant asymptote: $y = x + 1$. Vertical asymptote: $x = 1$. x-intercepts: $x = \pm 2$. As $x \to \pm\infty$, the graph hugs the line $y = x + 1$, while near $x = 1$ it shoots off to $\pm\infty$.

Full graph analysis: putting it all together

12 For $f(x) = \frac{2x^2-2}{x^2-9}$, find all intercepts, all asymptotes, and describe the symmetry of the graph (even, odd, or neither).

x-intercepts: $2x^2 - 2 = 0 \Rightarrow x = \pm 1$. y-intercept: $f(0) = \frac{-2}{-9} = \frac{2}{9}$. Vertical asymptotes: $x = \pm 3$. Horizontal asymptote (equal degrees): $y = 2$. Symmetry: $f(-x) = \frac{2x^2-2}{x^2-9} = f(x)$, so the function is EVEN, symmetric about the y-axis.

A compound rational inequality

13 Solve $\frac{x^2-1}{x-2} \leq 0$.

Factor numerator: $\frac{(x-1)(x+1)}{x-2} \leq 0$. Critical values $-1, 1, 2$. Sign chart: negative on $(-\infty, -1)$, positive on $(-1, 1)$, negative on $(1, 2)$, positive on $(2, \infty)$. Solution (including zeros, excluding undefined point): $(-\infty, -1] \cup [1, 2)$.

Graphing near a hole

14 For $f(x) = \frac{x^2 - 9}{x - 3}$, describe how the graph looks immediately around $x = 3$, and find the coordinates of the hole.

Simplify: $f(x) = x + 3$ for $x \neq 3$. The graph is just the line $y = x + 3$ with a single point removed at $x = 3$: the hole is at $(3, 6)$ — an open circle on an otherwise straight line.

15 Solve $\frac{3}{x+1} \geq \frac{2}{x-1}$.

Move all terms to one side: $\frac{3}{x+1} - \frac{2}{x-1} \geq 0 \Rightarrow \frac{3(x-1) - 2(x+1)}{(x+1)(x-1)} \geq 0 \Rightarrow \frac{x-5}{(x+1)(x-1)} \geq 0$. Critical values $-1, 1, 5$. Sign chart: negative on $(-\infty, -1)$, positive on $(-1, 1)$, negative on $(1, 5)$, positive on $(5, \infty)$. Solution: $(-1, 1) \cup [5, \infty)$.

16 A rectangular garden's cost per square meter to fence is modeled by $C(x) = \frac{200}{x} + 8x$, where x is the garden's width in meters. Find the vertical asymptote and horizontal/slant asymptote of this function, and interpret what the vertical asymptote means physically (why $x = 0$ isn't a realistic input).

Vertical asymptote: $x = 0$. As $x \to 0^+$, $C(x) \to \infty$ — a garden of near-zero width would require infinite fencing cost per unit length, which isn't physically meaningful; this confirms $x = 0$ must be excluded from the domain. There is no horizontal asymptote (the $8x$ term grows without bound), and the function behaves like $y = 8x$ for large x (a slant-like trend, though not a strict polynomial slant asymptote since $\frac{200}{x} \to 0$).

17 State the domain of $f(x) = \frac{\sqrt{x}}{x - 4}$.

Need $x \geq 0$ (for the square root) and $x \neq 4$ (denominator). Domain: $[0, 4) \cup (4, \infty)$.